

Swapnil Dhivare

 sdhivare.12@gmail.com  +91 9307066176  Mumbai  [LinkedIn](#)  <https://github.com/Swapnil-Dhivare>

Professional Summary

AI/ML and Data Science student with hands-on experience in machine learning, data preprocessing, model development, research-driven experimentation, and AI system engineering. Experienced in building RAG pipelines, multi-agent workflows (LangChain/LangGraph), visualizations, and deploying ML components across cloud platforms (GCP, AWS, Oracle Cloud). Passionate about applying AI to understand cognitive behavior, personalize learning systems, and drive data-backed insights. Skilled in Python, scikit-learn, Pandas, NumPy, TensorFlow/PyTorch (beginner), and scientific evaluation techniques.

Work Experience

AI Development Intern – Arkham Archives Jun 2025 – Sep 2025

- Engineered AI agents designed to function as personalized user guides by processing multimedia inputs (audio, text, media) to dynamically adapt and assign tasks using Morpheus AI, explored CAMEL for multi-agent system, worked with vector databases & LangChain.
- Implemented the ReAct pattern, enhancing agent reasoning and enabling more complex, multi-step problem-solving.
- Executed a seamless migration of application modules from GCP to AWS and Oracle Cloud, ensuring system resilience and cross-cloud scalability for the AI agent ecosystem.

WordPress Developer – Technoserve Engineers Aug 2024 – Oct 2024

- Spearheaded the company's digital transformation by developing their first-ever corporate website, establishing a critical online presence.
- Optimized website performance by 50% through strategic image compression and plugin management, improving user experience.

Project Contributor – Automated Course File Generation May 2024 – Jul 2024

- Automated academic documentation workflows using Python, reducing manual processing time for faculty by over 5 hours per week.
- Achieved a 95% reduction in manual data entry errors, significantly improving the accuracy and reliability of course outcome files.

Projects

Drishti AI - ISL Recognition System Present

- Developed a cutting-edge ISL recognition system leveraging facial and hand gesture analysis from video streams, aimed at achieving 75%+ accuracy to bridge communication gaps for the hearing impaired.
- Architected a ResNet50-based model using Flask, MediaPipe, and the INCLUDE dataset, achieving an initial accuracy of 70% with a target of 75%+.

AmbedkarGPT – Retrieval-Augmented Generation System Nov 2025

- Designed a modular RAG system for multi-document Q&A on Ambedkar's writings.
- Implemented ingestion, retrieval, and LLM-answering pipeline using LangChain + ChromaDB.
- Built a comprehensive evaluation suite: Hit Rate, MRR, Precision@K, ROUGE-L, BLEU, cosine similarity.
- Conducted experiments across chunk sizes and embedding strategies; identified optimal configurations and documented insights in a research-style analysis report.

Skills

Programming: Python, Java

ML & AI: scikit-learn, TensorFlow (basic), PyTorch (basic), CNNs, RAG, embeddings, evaluation metrics

Cloud: AWS, GCP, Oracle Cloud

Data Analysis: Pandas, NumPy, Matplotlib, Seaborn, Data Cleaning, Feature Engineering, Statistical Analysis

Agent Systems: LangChain, LangGraph, Multi-Agent Architecture, ReAct pattern

Visualization: Matplotlib, Seaborn, Looker Studio (familiar), Dashboards, Reporting

Tools: Git, Docker, SQL, Postman, Streamlit, Flask

Soft Skills: Research documentation, experiment design, report writing, teamwork, presentation

Education

Smt. Indira Gandhi College of Engineering

Bachelor of Engineering in Computer Science

Relevant Courses: Machine Learning, Data Science, Web Technologies

Volunteering & Leadership

Team Lead – Smart India Hackathon

- Led a team to win the college-level Smart India Hackathon by conceptualizing and developing a specialized language translator for military applications using Large Language Models (LLMs).

Literature Tutor – Self-Employed

Nov 2023 – Nov 2024 | Mumbai

- Provided part-time tutoring in literature to high school students, enhancing their understanding and academic performance over nearly one year.

Certifications

Machine Learning Specialization – Stanford University & DeepLearning.AI (Coursera)

CS50x: Introduction to Computer Science – Harvard University

Technoserve Engineers Certificate

Arkham Archives Certificate